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A Spelunky Game Design Analysis - Pt. 2

The second part of my Spelunky game design analysis, addressing the topics of randomness, fairness, control, skill, strategy, and scaffolding.

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November 15, 2012

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[Here's Part 1 of the analysis.]

Randomness

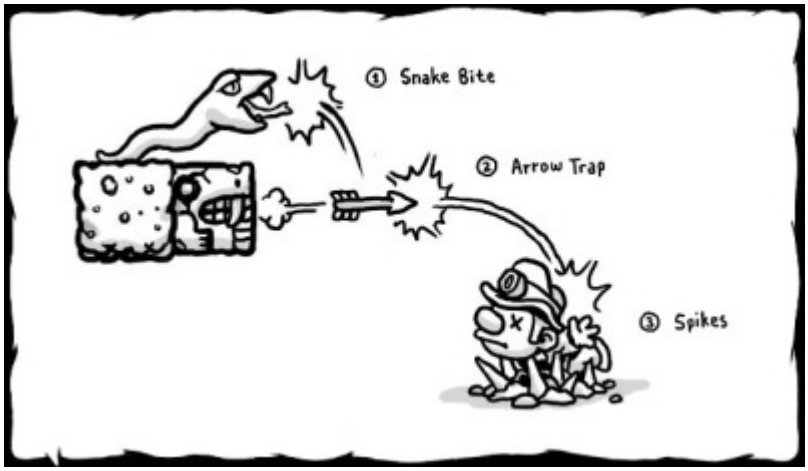
I've said it before, and it's worth stating again; I am a fan of randomness in game design. When used well, randomness can help give gameplay actions a clear risk-reward balance. Random elements can also stress adaptation skills in a unique way. And randomly generated levels are a core element of the roguelike genre. Every level you play in Spelunky is unique. And when you die and restart you'll never see the same level again. Derek Yu describes playing randomly-generated levels as "fresh" and "exciting" where "you'll have to think on your feet." The idea with all roguelikes is that the player must constantly adapt because there's little to memorize.

> **"With Spelunky, you are never learning a 'piece' of music.... It's still a game about repetition and learning, but what you are learning is the overall composition, understanding the overall system and how it works, and becoming fluent in that." ~IGF Judge**

Another way of explaining the above quote is because Spelunky is randomly generated and players cannot get repeated practice on any particular challenge, they're force to learn the general rules of the system. In this case, the general rules that govern each environment and level theme; e.g. the "Snakes, I hate snakes!" levels always feature a snake pit. In other words, the skill ceiling is lowered so that thehigh levels of strategy, memorization, and long term planning are closer to the low levels of improvisation, and tactical play.

Because the levels are randomly generated in Spelunky, highly polished, layered gameplay challenges are not likely (see Super Mario Bros for examples). Even if a level in Spelunky was randomly generated with the layered design and polish of a Mario level, players would only be able to experience it once before the level becomes a ghost of history, never to be experienced again. Furthermore, to best appreciate and understand layered or deep design, such levels must be played multiple times. Clearly, the Spelunky levels aren't designed to be played and appreciated like layered Mario levels, which is a good thing considering its roguelike design features. But there's another, more serious drawback randomly generated levels have on the overall gameplay experience.

Generally for all games players seek to exert as much control as they can over level challenges in order to win or achieve some other level of success. We all start this control exerting, power grabbing, skill building process the same way; by learning the complexities (rules) of the game. With this knowledge we can make better informed decisions and weigh out the risk versus reward of our options. **With Spelunky, the bottom line is, when faced with unpredictable, deadly challenge everywhere, progressing slowly, overcoming every obstacle step-by-step is the safest, and, in my experience, the only way to play successfully.** While this approach is effective, it reduces the entire gameplay experience to a series of simple, individual, interactions.



I'm pretty sure this happened to me... more than once. ~source

I find that when I freestyle in Spelunky, I die very quickly. And I can't see how the results would be any different for anyone else. With the levels being randomly generated, you simply don't know what's around any corner. And in a game where you can instantly die from one wrong jump or be comboed to death by the lowliest of enemies (see cartoon above), throwing caution into the wind is ultimately throwing your life away. You may get lucky every now and then, but luck can only get you so far in such a skill based game. There is simply much more risk than reward in nearly every inch of the game.

> **"Spelunky looks like a game of execution, but it's really a game about information and decision-making. How good are you at looking at a situation and understanding what it means? You can't memorize... you must rely on your literacy of the system..." ~IGF Judge**

I ultimately consider Spelunky a roguelike first and a platformer second. To play the game well, players must play with great caution. Even high level play consists of an accelerated version of this cautious approach (see video1, video2, or video3). Spelunky is most interesting when we consider how the moment to moment decisions fit in the larger context of beating the game and overcoming great odds. Still the platforming gameplay moment to moment isn't so interesting. Ultimately, roguelike gameplay is a fight for control against staggering odds, which brings us to the topic of fairness.

Fairness and Control

In a game so deadly and so random, it's good to be able to rely on something. It's important to consider controller design and mechanics as the foundation of player control and skill. If the controls are inconsistent, overly complex, or confusing, the rest of the gameplay experience rests on a questionable base.

In Spelunky I've found a few odd design choices and controller issues. I can understand why players automatically drop their held items after a few seconds of teetering on the ledge. But what I don't understand is why this occasionally happens when I run off the a ledge without teetering (see close example here). I also don't understand why I'll sometimes swing around and grab a ledge when I was running forward at full speed. Normally this only happens when you crawl over a ledge. Also, I don't like how characters automatically grab ledges. It would be nice to be able to hold down and not grab the ledge like in Smash Brothers especially considering how easy it is to run into a ledge because of the high character movement speed and how easy it is to confuse the grabbing and standing state because of the lacking sound design. While these quirks are odd, I don't consider them significant enough to make the game unfair.

As you adventure in Spelunky, some levels will randomly be plunged into darkness (see video here). It could be a cave level, a jungle level (including the Black Market), or a temple level. Fortunately, at the start of each dark level is a torch you can carry to gain visibility and light other torches with. The dark levels in Spelunky PC (see video here) are fair compared to the unfair XLBA version of Spelunky for three reasons. 1) Players get access to multiple flares/torches. 2) The flare/torch light is much larger and brighter illuminating more. 3) Hazards and other enemies don't move when off the screen so it's less likely for anything off screen in the darkness to harm you.

Though I like the idea of using the single torch to light lamps in the new Spelunky, it's just not enough light. Carrying the torch also comes at the cost of dropping any held item you may be carrying. In my best run of Spelunky I made it to the temple with everything I needed to enter the City of Gold, and it was a dark level. By some miracle I was able to safely find the golden door and escape just before the ghost touched me. It's hard enough to deal with some challenges in the light. It's clear to me that some level combinations plus darkness are just a roll of the dice for survival. For example, arrow traps on the ground level are extremely cheap in dark levels (see close example). You can't see the trap coming. And if you're holding a torch, you can't panic whip for safety either. And considering how enemies can see you just fine in the darkness when you can't see them, the new dark level design is the a major unfair element.

Skill and Strategy

Overall, I feel that there is only one consistent way to exert skill and take control of your Spelunky destiny; **taking it slow and leveraging the extreme, undermining powerups via the guaranteed (predictable) level elements.** Yes, you can beat the game without grabbing any items or powerups (or gold for that matter) just like you can beat Super Mario Brothers as Small Mario. But with Spelunky, you're bound to run into a dark level, a colony of giant spiders, a Mummy Anubis combo, or some other challenge that is simply too risky to take on with just your whip and your wits. At these times, you'll want to use a powerup, which is by nature a design element that allows the player to undermine challenge in some way. The bottom line is, the more powerups you have in Spelunky, the less risky the game becomes. Many powerups can be stacked allowing players to become super powerful. And there's only one "true" way to get as much power as possible.

The left guy loves platforming. No ropes. No Bombs. No Problem.

The right guy is powered up and ready to play Spelunky. ~source

Understanding the one "true" way to play Spelunky becomes clear when you consider the goal of the game and the steps it takes to reach it. To complete each level you need to navigate downward and find the door. The entire level is destructible so bombs becomes the most precious resource for combat and navigation. Bombs are limited in supply. Other than getting lucky with random item drops, the shop keepers are a reliable source of bombs and other items. Since these shop keeps charge so much money for their items, it's far more advantageous to rob them. Sure, this makes you wanted for the rest of the game, but with enough powerups you'll be able to handle the shopkeepers with ease. Once you rob one shopkeeper, you might as well use that power to rob them all. With ropes, climbing gloves, or a jet pack you can travel just about anywhere safely. And after killing the shopkeeper, you'll probably have a shotgun, which is an extremely powerful, long range weapon with unlimited ammo.

At some point in the cave levels, there is guaranteed to be an Udjat Eye locked in chest. The key is always on the same floor as the chest. With the Udjat Eye, you can easily locate the black market on one of the jungle levels. This hidden location can only be reached with bombs (which you should have plenty of by now). There is always a black market in one of the jungle levels. In the black market, you should take everything, but your priority is the Ank. After looting the black market you must locate the Moai head statue that's guaranteed to be on one of the ice levels. Then kill yourself near the statue. The Ank will revive you and you'll receive the Hedjet head gear. From there you should have plenty of undermining power to take out Anubis and grab his golden sceptre. Use the Sceptre to enter the City of gold... etc.

> **"And even though the game is randomized, players of the original enjoyed highscoring and speedrunning, so we're trying to limit the luck involved." Derek Yu**

The point is, though you don't have to do all of these steps to be successful, they are the safest and more reliable strategy. After all, strong strategies are built around known factors and minimal risk. Because the steps I described involve many guaranteed elements, you can rely on and perfect the strategy. This is how Derek Yu decided to minimize the luck involved in high level play. Just by raiding the black market you should plenty of equipment and powerups to make the rest of the game fairly easy by undermining everything with explosives and gravity defying power. And until you learn this one "true" strategy or devise a similar strategy, you'll largely be rolling the dice with your efforts.

Scaffolding

According to the one "true" way to play Spelunky, the biggest hurdle you have to conquer Spelunky is one of acquiring knowledge not honing execution. Pro platforming execution skills may get you out of tight spots, but ultimately planning ahead and leveraging powerups keep you from getting into these tight spots in the first place. There are a few design features in Spelunky I want to highlight that are especially helpful in creating learning environments that help players learn much faster and more effectively; i.e. **scaffolding**.

I love the new clued secrets design in Spelunky. There are serveral little feedback touches that make finding some secrets more clear. The Eye of Udjat pings more rapidly when you drawn near the black market. The black market is always hidden behind a distinct bunch of leaves. The book of the dead writhes when you're positioned above the secret entrance. And the door to the city of gold is no longer hidden behind things or at the bottom of lava pools. The cleaner clue design helps players to think and wonder about the mysteries beyond the obvious. Designing secrets to be hidden but not completely obtuse is an art in itself, and Spelunky strikes a better balance than its PC predecessor.

I also like how collecting items or killing enemies unlocks their individual journal entries. Unfortunately, these entries are closer to flavor text than gameplay tips. Though I'm not a fan of level editors in general, I think that Spelunky could have really benefited from a level editor as a scaffolding tool like the PC version has. It would be neat if once you kill an enemy or obtain an item, you unlock it in the journal and in the level editor. In the editor you would be free to create scenarios to practice. This feature would work a lot like how Mega Man 10's challenge mode let players target practice specific bosses to build their skills risk free. Or like Bangai-O Spirits's level editor that let's player alter and practice any stage at any time.

Consistent exposure and practice are essential when learning anything. We simply learn better when we focus on a task and apply ourselves for 15-50 minutes at a time. To learn we often conduct many experiments (trial-and-error), which require many repetitions under the same conditions. **One negative effect randomized level design has on learning is that the unpredictable, inconsistent challenge scrambles the learning process.** When you make mistakes in games, as you inevitably will, you will form questions and adjust your approach for the next attempt. The sooner you can test out your new approach the better. After all, holding these questions and notes in your head for a long time takes up brain space and increases the likelihood of forgetting something. Naturally, the random level design and the permadeath of Spelunky make it more difficult to learn in this traditional way.

Fortunately, Spelunky is designed with a clever short cutting system. By doing various tasks, players can unlock shortcuts that allow them to skip levels, starting their adventure on the jungle, ice, or temple level. This is great for getting some focused practice on the elements and dangers of these areas. Without worrying about losing a lot of progress and time from dying, players can really let loose. Ultimately, to have the best odds at success and earn the most money, players must play from the beginning of the game. But as far as learning goes, these short cuts are great.

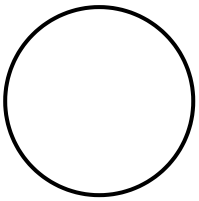
In closing, the new Spelunky is a superior version of the original game, which I foolishly never put on my GOTY list for 2009. I won't make that mistake again for 2012. The game is packed with detail, dynamics, and interplay. It's a bit janky, a bit too fast, a bit unfair at times. But overall, it's still a very solid game, like the PC version. Just like any skill based game, the first challenge is to build up your skills. Don't use the randomness and the difficulty as an excuse. Yes, the constantly shifting walls makes the game harder to learn, but there are many ways to build your knowledge skills these days. Watching youtube and sharing stories with each other is a legitimate way to scaffold your learning process. And co-op is a fantastic teaching tool when you play with a more skilled player or when you let a friend go first and learn from their mistakes. While there may be a few "true" strategies for obtaining big money with the least amount of risk, there are certainly many ways to play and enjoy Spelunky. Particularly, the achievements are designed to encourage players to attempt these various types of challenges and experiences.

Spelunky is game design gold. With just a bit of cash and effort, you too can participate in this alchemic dream where every attempt at success is different yet familiar. Oh, so familiar.

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